

TRAN VIET KHOA

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EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE (NUS)
Bachelor of Engineering, Computer Engineering

Aug 2024- May 2028

PROJECTS

KForum, Full Stack Developer

Nov 2025 - Now

Github Link: <https://github.com/Michael-coding06/KForum.git>

- Developed a **high-concurrency** web forum utilizing **horizontal scaling** and **load balancing** to ensure low-latency performance and scalability for a growing user base.
- Architected a containerized full-stack application using **Docker and AWS EC2**, designing a secure PostgreSQL schema and **JWT-based authentication** with **HTTP-only cookie** handling.
- Engineered a bi-directional messaging layer using **Socket IO** and **Redis Pub/Sub** for real-time commenting and implemented automated Cron jobs for scheduled database backups to ensure data persistence.
- Tech Stacks: Go (Gin), React, TypeScript, REST API, PostgreSQL, Docker, AWS EC2, Socket IO, Redis, Material UI (MUI).

NUS Schedule Boss - Orbital 2025 Apollo 11, Full Stack Developer

May 2025 – Jul 2025

Github Link: <https://github.com/Michael-coding06/NUS-ScheduleBoss>

- Developed an **AI-integrated** module management platform to streamline academic planning through NLP and interactive data visualization.
- Integrated an **OpenAI**-powered chatbot with persistent MongoDB history to automate scheduling via NLP and aggregated **Reddit-based course reviews** for informed decision-making.
- Engineered a full-stack solution featuring a dynamic timetable visualizer and a contextual notification system to provide with personalized, real-time schedule reminders.
- Tech Stacks: React, Node.js, Express.js, MongoDB, OpenAI API, Supabase, Reddit API

LEADERSHIP EXPERIENCE

Vietnamese Youth Alliance in Singapore, Web Developer & Logistics Coordinator

Oct 2024 - Sep 2025

- Led "Singapore Study Dream 2024" seminar (150+ attendees), partnering with 8 top universities and connecting 50 mentees with 40 mentors.
- Coordinated Career Fair 2025 logistics, engaging 100+ Vietnamese students in Singapore.
- Built user-friendly landing pages for major events, improving participant engagement and streamlining event coordination processes.
- Organisation's website live at: [Website](#).

ADDITIONAL SKILLS

- Programming: Go (Intermediate), C++ (Intermediate), Python (Intermediate), JavaScript (Intermediate), Java (Beginner)
- Web Development: React, Node.js, Express, MongoDB, HTML/CSS, Docker, EC2
- AI & APIs: open ai API, TensorFlow, PyTorch, Keras, Pandas, NumPy, Matplotlib
- Technical Knowledge: Full-stack development, database design, agent-based AI
- Languages: Vietnamese (Native), English (Fluent), Mandarin (Intermediate)